IT Project Management

Lecture 5:
Project **Schedule** Management

—Planning & Controlling Time—

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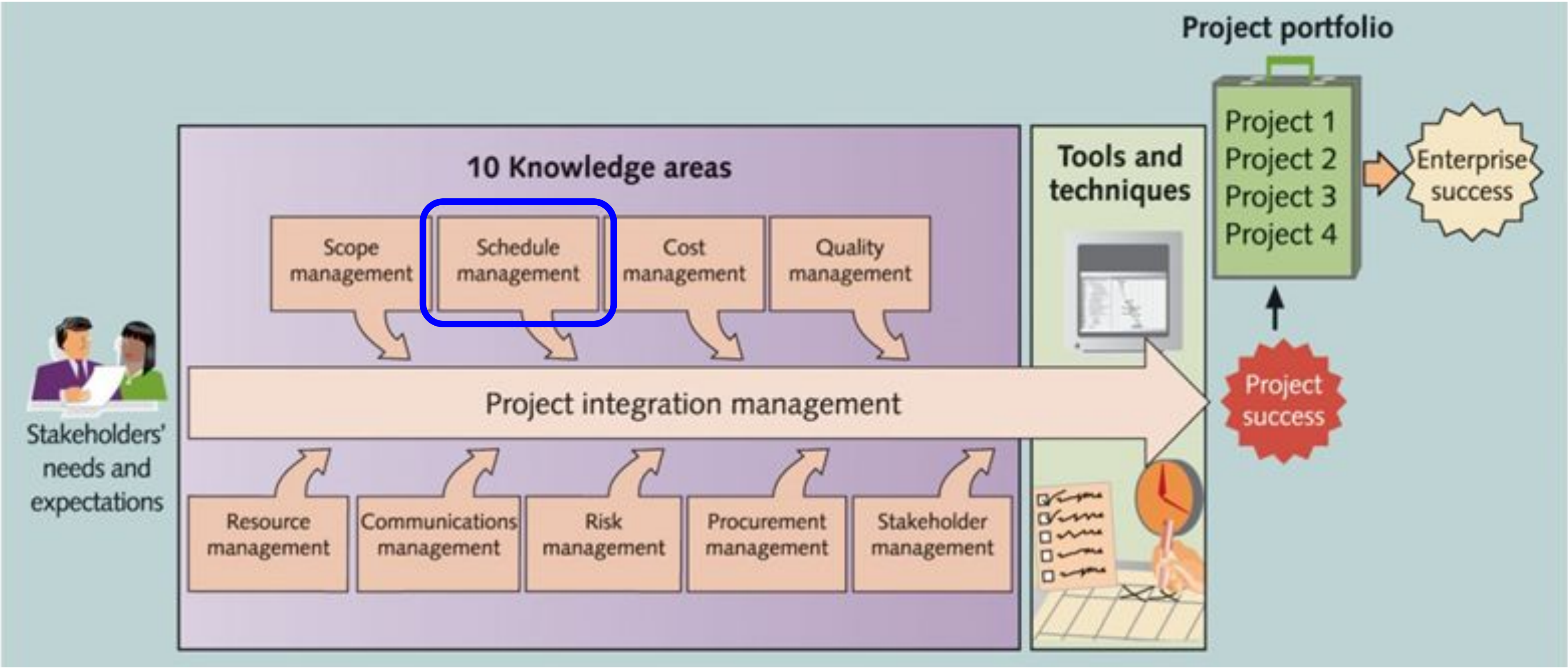
2025-2026

Project **Schedule** Management



About this lecture

📌 Learning objectives:

- **Understand project scheduling processes** → Learn how activities are planned, sequenced, and tracked
- **Explore key scheduling tools (CPM, CCM, PERT, Gantt, etc.)** → Learn how each method works with real-world examples
- **Apply time estimation & optimization techniques** → Understand how to estimate task durations accurately and shorten schedules efficiently

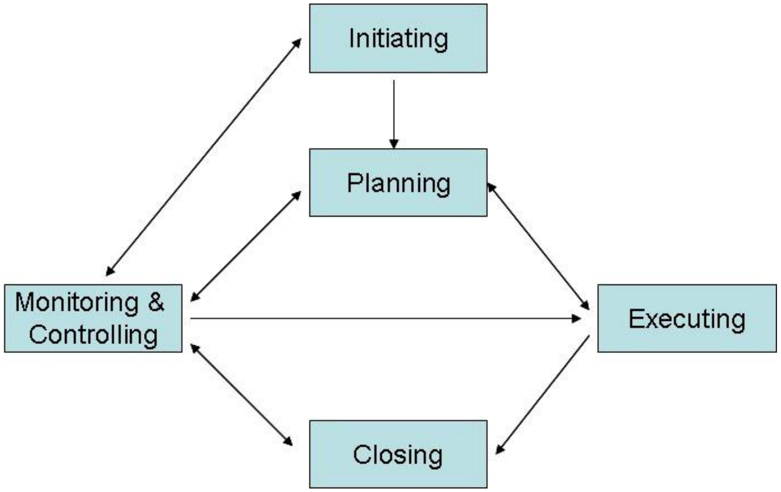
Introduction to Project Scheduling

What is Project Scheduling?

Project scheduling is the process of planning and managing the timeline of tasks to ensure timely completion

→ Defines when tasks should **start and finish**, who is **responsible**, and how tasks **depend on** each other

→ Helps manage **resources, risks, and deadlines** efficiently



Why is Project Scheduling Important?

Key Benefits of Scheduling:

- Prevents Missed **Deadlines** → Ensures tasks are completed on time
- Controls **Project Budget** → Avoids overspending due to poor time planning
- Improves **Resource Allocation** → Prevents bottlenecks & overallocation
- Enhances **Communication** → Keeps stakeholders informed of progress
- Identifies **Risks Early** → Allows for better contingency (backup) planning

Challenges in Project Scheduling

Time is the least flexible constraint: It moves forward no matter what!

CHAOS Report on IT Project Delays:

- In 1995, the average project delay was **222%**
- By 2001, it improved to **163%**, but **delays remain common**

🎤 **Question:** *Why do most projects fail to meet their deadlines?*

Challenges in Project Scheduling

Why Do Projects Miss Deadlines?

- **Poor Initial Planning** → Unclear goals & underestimation of time
- **Scope Creep** → New requirements cause schedule delays
- **Resource Constraints** → Limited personnel or funding
- **Task Dependencies** → Delays in one task affect the entire project
- **Lack of Communication** → Delays in decision-making and approvals
- **Unexpected Risks** → Technical issues, external dependencies, etc.

Challenges in Project Scheduling

How to Overcome These Challenges?

1. Use **scheduling tools** and techniques to estimate time
2. Add **schedule buffers** to absorb unexpected delays
3. Regular **progress tracking** to **adjust schedules** proactively

Project Scheduling Process

The 5 Key Steps of Project Scheduling Process

1. **Define Activities**: Identify and break down work into tasks
2. **Sequence Activities**: Determine dependencies between tasks
3. **Estimate Durations**: Predict time required using techniques like PERT
4. **Develop the Schedule**: Create a timeline using tools like Gantt charts
5. **Control the Schedule**: Monitor progress and make adjustments

Step 1: Define Activities

What is **Activity Definition?**

- Lay the **foundation** for developing the **project schedule**
- **Identify all tasks** required to complete the project
- Use **Work Breakdown Structure (WBS)** to break down tasks
- Help in **accurate** planning, estimation, and execution

Step 1: Define Activities

Key Documents in Defining Activities:

- **Project Charter**:
  - Contains the **start & end dates**
  - Includes **budget details**
- **Scope Statement & Work Breakdown Structure (WBS):**
  - Defines **what needs to be done** in the project
  - Includes **all tasks** and ensures all tasks are accounted for

Step 2: Sequence Activities

What is **Activity Sequencing?**

- Determine the **order of tasks** and their **dependencies**
- Typically visual use tools like **network diagrams**
- Essential for using scheduling techniques like **Critical Path Method**

Step 2: Sequence Activities

Three Types of Dependencies in Scheduling:

1. **Mandatory Dependency (Hard Logic)**
   - Fixed by the nature of the work
     - E.g., Walls must be built before painting
2. **Discretionary Dependency (Soft Logic)**
   - Determined by the project team
     - E.g., Testing before a second design review
3. **External Dependency**
   - Involves factors outside the project
     - E.g., Waiting for government approval

Step 2: Sequence Activities

Time Constraints in Scheduling:

- **No Earlier Than:** Task **cannot start before a certain date**
- **No Later Than:** Task **must be completed before a deadline**
- **On This Date:** Task must be completed **exactly on a fixed date**

Other Types of Constraint:

- **Management Constraints:** Set by **project managers** for business needs
- **Technical Constraints:** Hardware, software, or **infrastructure limitations**
- **Resource Constraints:** Limited availability of **team members or equipment**
- **Organizational Constraints:** Company policies, **funding limits**, or approvals

Step 3: Estimate Durations

Predict **how long each task will take**, with **3 methods**:

- **Expert Judgment**

  → Use human experts for estimation

- **Historical Data**

  → Use past performance of similar projects for estimation

- **PERT (Program Evaluation & Review Technique)**

  → Use **three-point estimate**: optimistic, pessimistic, and most-likely estimates

Step 4: Develop the Schedule

- Use scheduling tools like **Gantt Charts, Critical Path Method (CPM), and Critical Chain (CCPM)**
- Identify **project milestones** and deadlines
- Allocate **resources efficiently** to avoid overloading team members

Step 4: Develop the Schedule

What are milestones?

A **milestone** is a **significant event** in the project

- It often takes several activities and **a lot of work to complete** a milestone
- They're useful tools for **setting schedule goals** and **monitoring progress**

Examples:

- Obtaining customer sign-off on key documents
- Completion of a specific product

SMART Criteria for milestones:

- Specific
- Measurable
- Assignable
- Realistic
- Time-framed

Step 5: Control the Schedule

- Regularly compare **actual progress vs. planned schedule**
- Use **Earned Value Management (EVM)** to measure progress
- Implement **schedule compression techniques (Fast Tracking, Crashing)** to recover lost time

Exercise: Plan a University Seminar Schedule

📌 **Instructions:**

1️⃣ **List key tasks required to organize a university seminar.**

2️⃣ **Sequence the tasks in logical order.**

3️⃣ **Estimate the time needed for each task.**

4️⃣ **Identify dependencies between tasks.**

Exercise: Plan a University Seminar Schedule

Example:

- ✔ **Task 1:** Define seminar topic and objectives.
- ✔ **Task 2:** Identify and invite speakers (dependent on Task 1).
- ✔ **Task 3:** Book venue (parallel task, but must be completed before promoting).
- ✔ **Task 4:** Promote the seminar and open registrations (dependent on Task 2, 3).
- ✔ **Task 5:** Conduct the seminar (dependent on Task 4).
- ✔ **Task 6:** Gather feedback & report results (dependent on Task 5).

Project Scheduling Techniques

PERT: Program Evaluation & Review Technique

What is PERT?

- A **statistical tool** used in project management for **estimation**
- Developed by the **U.S. Navy (1950s) for complex projects**
- Helps **predict task duration** by considering **uncertainty**

→ Considering **best-case, worst-case, and typical case scenarios**
→ Create **more realistic plan** and **reduces schedule risks**

PERT: Program Evaluation & Review Technique

PERT Formula:

$$ET = \frac{O + 4M + P}{6}$$

where:

- **O (Optimistic)** → Best-case scenario (fastest completion)
- **M (Most Likely)** → Expected case (normal completion)
- **P (Pessimistic)** → Worst-case scenario (longest delay)

PERT: Program Evaluation & Review Technique

Exercise: Estimating Software Module Completion Time

- Optimistic Time (O): 5 days
- Most Likely Time (M): 10 days
- Pessimistic Time (P): 20 days

PERT: Program Evaluation & Review Technique

Exercise: Estimating Software Module Completion Time

- Optimistic Time (O): 5 days
- Most Likely Time (M): 10 days
- Pessimistic Time (P): 20 days

$$ET = \frac{5 + 4(10) + 20}{6} = \frac{5 + 40 + 20}{6} = \frac{65}{6} = 10.83 \text{ days}$$

Gantt Charts

What is a Gantt Chart?

- A **visual scheduling tool** that represents tasks on a **timeline**
- Developed by **Henry Gantt** in the early 1900s
- Shows **task duration, start & end dates, and dependencies**

→ **Clear visualization** of the entire project timeline

→ Helps teams **understand task dependencies**

→ **Improves communication** between team members

→ Helps track **progress** and manage **workloads efficiently**

Gantt Charts

How to Read a Gantt Chart:

- **Tasks listed on the vertical axis**
- **Timeline on the horizontal axis**
- **Bars represent tasks**, showing **start & end dates**
- **Arrows indicate dependencies** between tasks

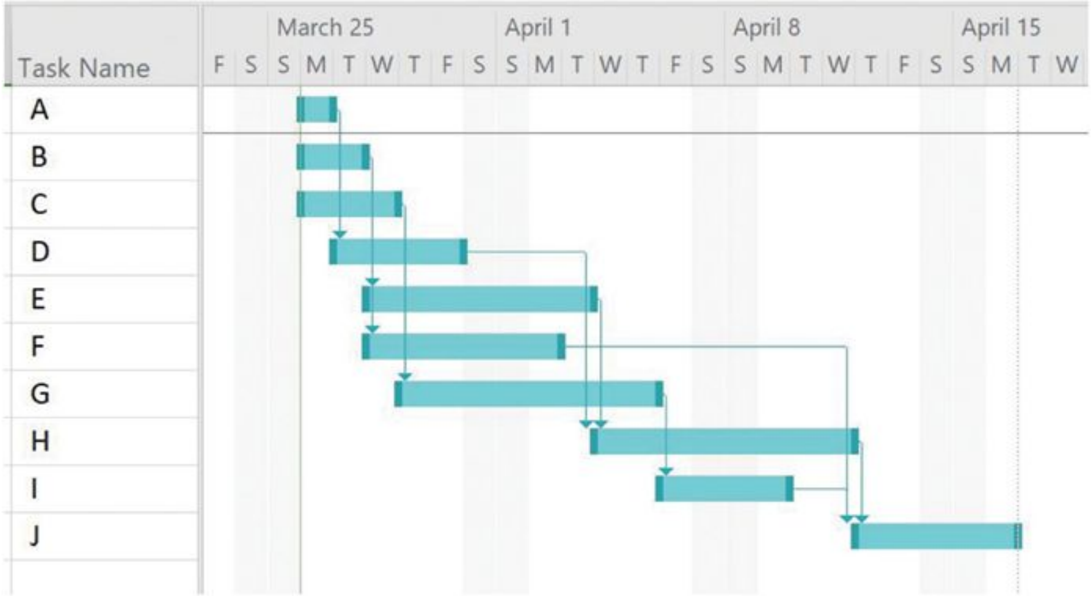


FIGURE 6-5 Gantt chart for project X

Gantt Charts: Example

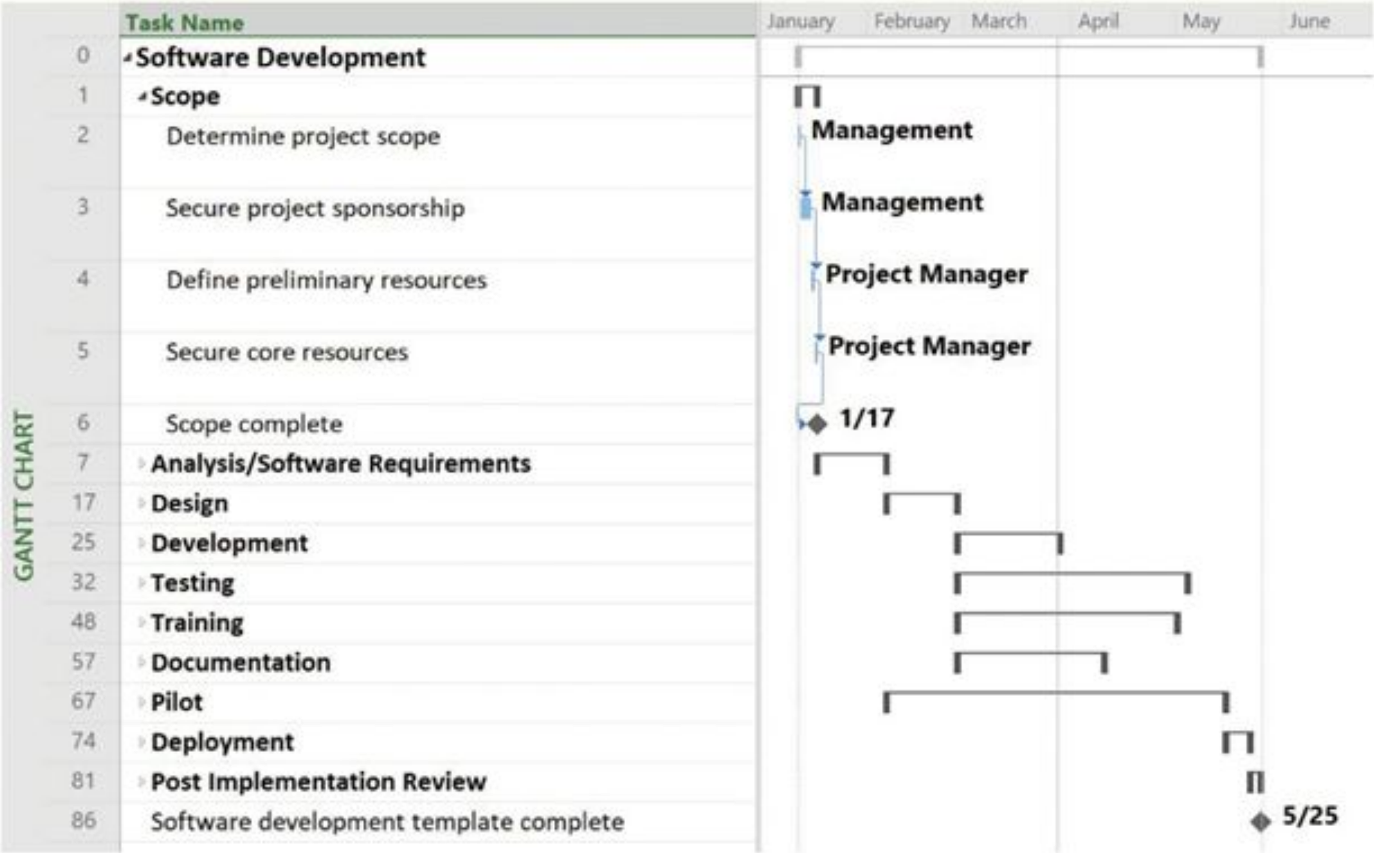


FIGURE 6-6 Gantt chart for software launch project

Gantt Charts: Example

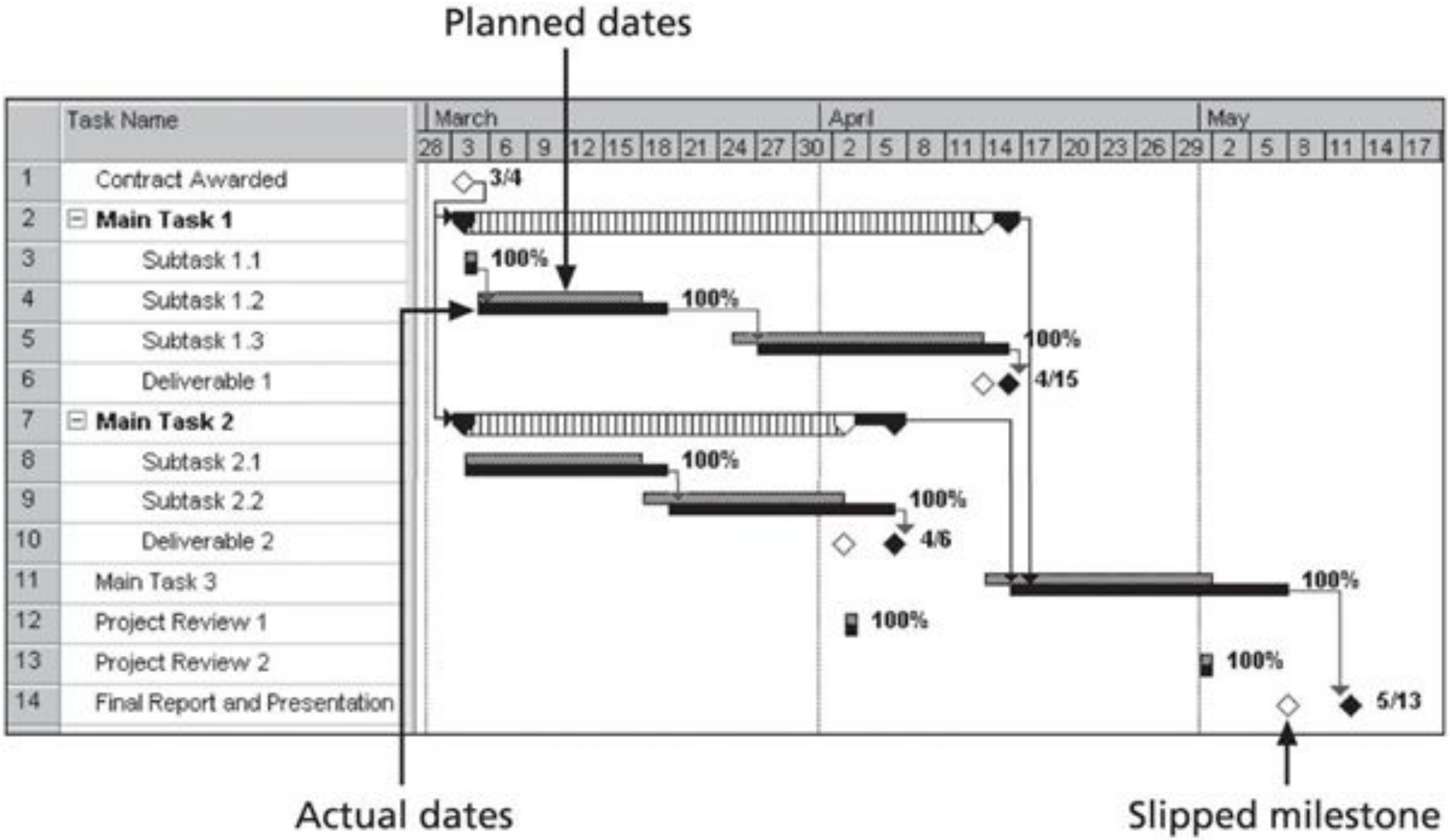


FIGURE 6-7 Sample tracking Gantt chart

Gantt Charts

Exercise: Create a simple Gantt chart

Create a simple Gantt chart for an IT project (e.g., building a mobile app)

- **Task 1:** Requirements Gathering (Week 1-2)
- **Task 2:** UI/UX Design (Week 3-4) (Depends on Task 1)
- **Task 3:** Coding Backend (Week 5-8) (Depends on Task 2)
- **Task 4:** Coding Frontend (Week 5-7) (Depends on Task 2)
- **Task 5:** Testing (Week 9-10) (Depends on Task 3, 4)
- **Task 6:** Deployment (Week 11) (Final step)

Network Diagrams

Network Diagrams: a useful tool to visualize the order of activities

Displays the **logical relationships** between tasks in a **flowchart format**

→ Shows **which tasks depend on others**

→ Helps determine the **Critical Path (longest sequence of tasks)**

→ Used in **Critical Path Method (CPM) analysis**

Network Diagrams

Two main formats:

- **AOA**: Activity-on-Arrow
    - Activities are represented by arrows
    - Nodes are the starting and ending points of activities
    - Each node is a **state** of the project, e.g., start, finish, etc.
- **AON**: Activity-on-Node
    - Activities are represented by nodes
    - Dependencies are represented by arrows
    - Also called PDM (Precedence Diagramming Methods)

Network Diagrams

Note: understand 4 types of dependencies in AON/PDM

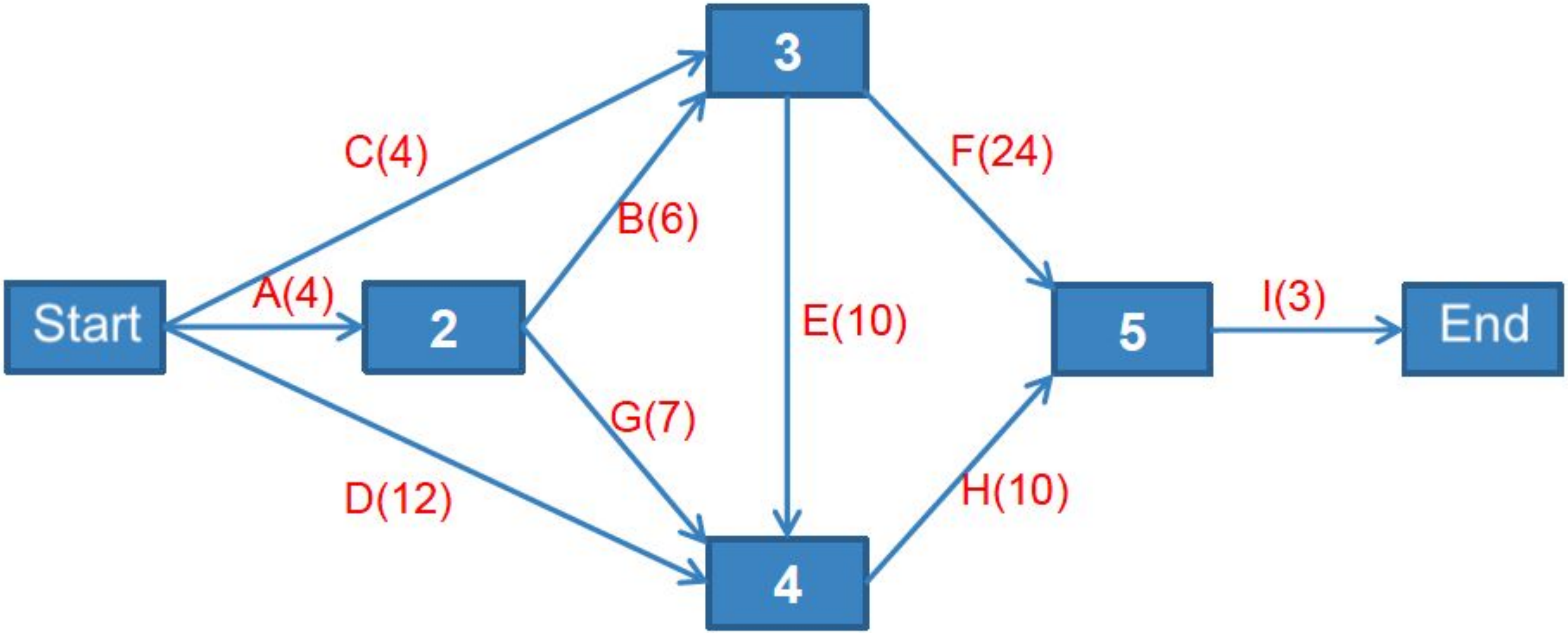
Dependency Type	Description	Example
Finish-to-Start (FS)	Task B starts only after Task A finishes.	Coding must finish before testing can begin.
Start-to-Start (SS)	Task B starts at the same time as Task A.	Frontend and Backend development start simultaneously.
Finish-to-Finish (FF)	Task B finishes only when Task A is completed.	Writing documentation and software testing must finish together.
Start-to-Finish (SF)	Task B cannot finish until Task A starts.	A security system must be running before the old system is shut down.

Network Diagrams: **Example**

Task	Predecessor Task	Duration
A	-	4
B	A	6
C	-	4
D	-	12
E	B, C	10
F	B, C	24
G	A	7
H	D, E, G	10
I	F, H	3

Network Diagrams: **Example**

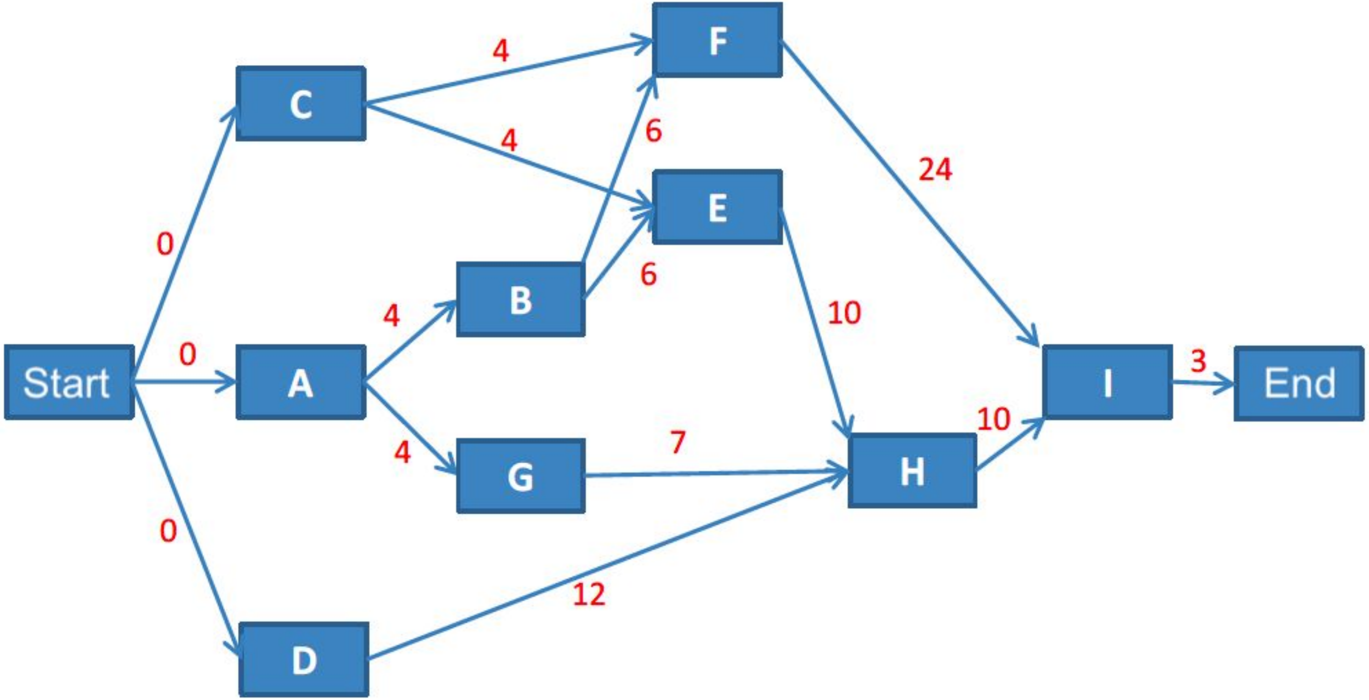
Activity on Arrow (AOA)



Network Diagrams: **Example**

Activity on Node (AON), aka PDM

Numbers on the arrows are durations → Use this for Critical Path Analysis

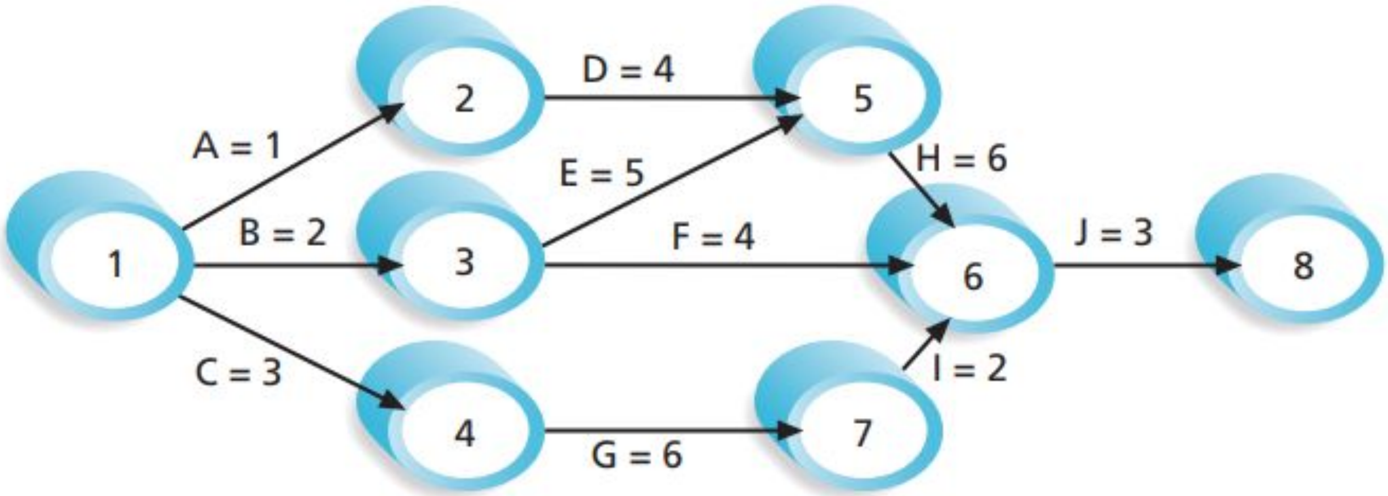


Network Diagrams: **Example (other notations)**

Task	Predecessor Task	Duration
A	-	1
B	-	2
C	-	3
D	A	4
E	B	5
F	B	4
G	C	6
H	D, E	6
I	G	2
J	F, H, I	3

Network Diagrams: **Example (other notations)**

Activity on Arrow (AOA)



Note: Assume all durations are in days; A=1 means Activity A has a duration of 1 day.

FIGURE 6-2 Activity-on-arrow (AOA) network diagram for Project X

Network Diagrams: **Example (other notations)**

Activity on Node (AON), aka PDM

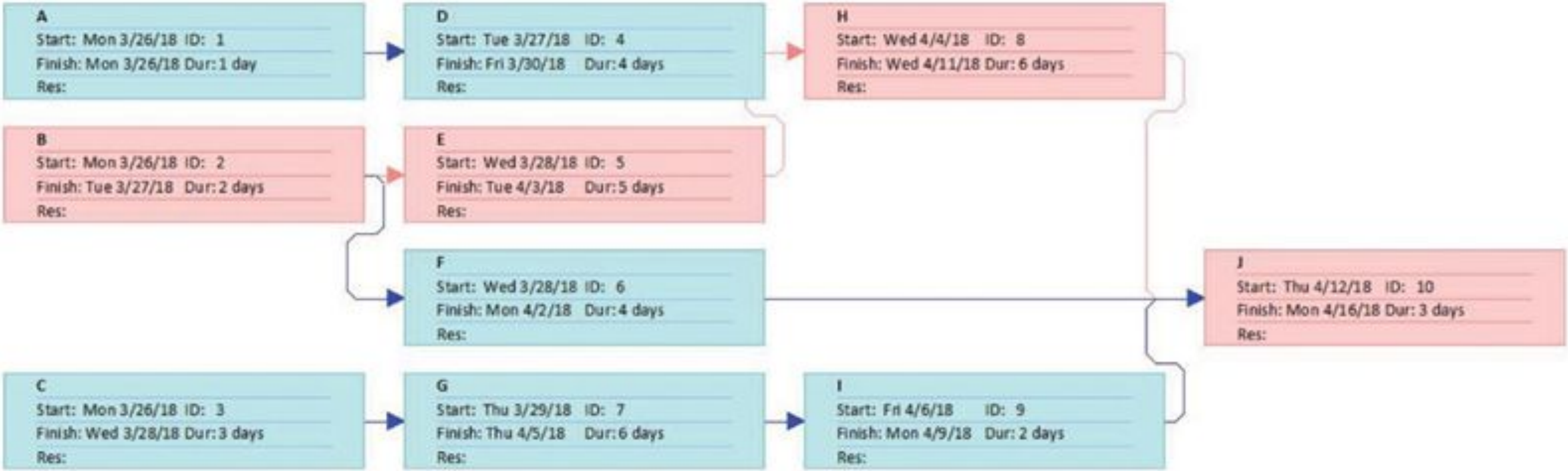


FIGURE 6-4 Precedence diagramming methods (PDM) network diagram for project X

Critical Path Method (CPM)

What is the Critical Path Method (CPM)?

- **The Critical Path** is the **longest sequence of dependent tasks** in the project

  → Calculates the **minimum project duration** (otherwise some tasks cannot finish)

- If **any task in the Critical Path is delayed, the entire project is delayed**

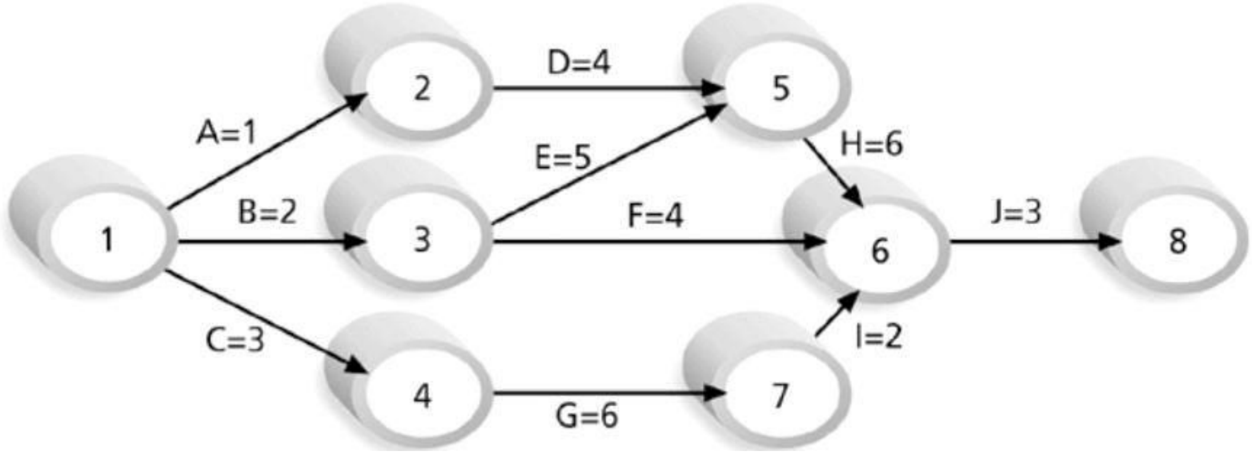
  → Identifies critical tasks with no flexibility (zero slack)

- Helps **identify tasks that need the most attention** to keep the project on track

  → Improves time & resource management

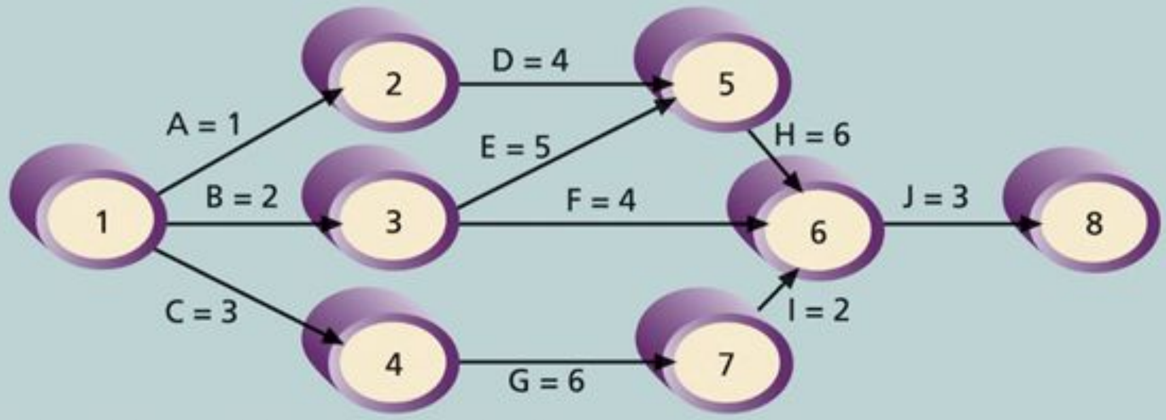
Critical Path Method (CPM)

Example: Finding the Critical Path in a Project



Critical Path Method (CPM)

Example: Finding the Critical Path in a Project



Note: Assume all durations are in days.

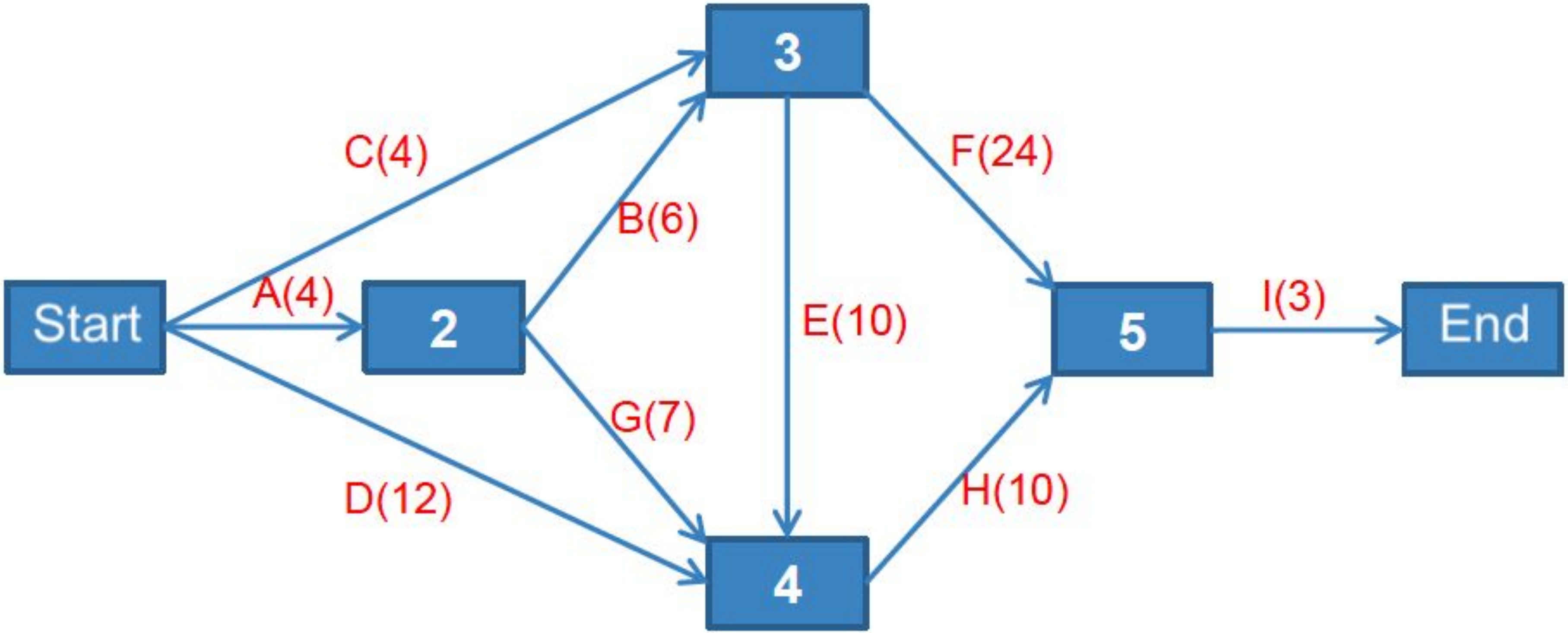
Path 1:	A-D-H-J	Length = 1+4+6+3 = 14 days
Path 2:	B-E-H-J	Length = 2+5+6+3 = 16 days
Path 3:	B-F-J	Length = 2+4+3 = 9 days
Path 4:	C-G-I-J	Length = 3+6+2+3 = 14 days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

FIGURE 6-8 Determining the critical path for project X

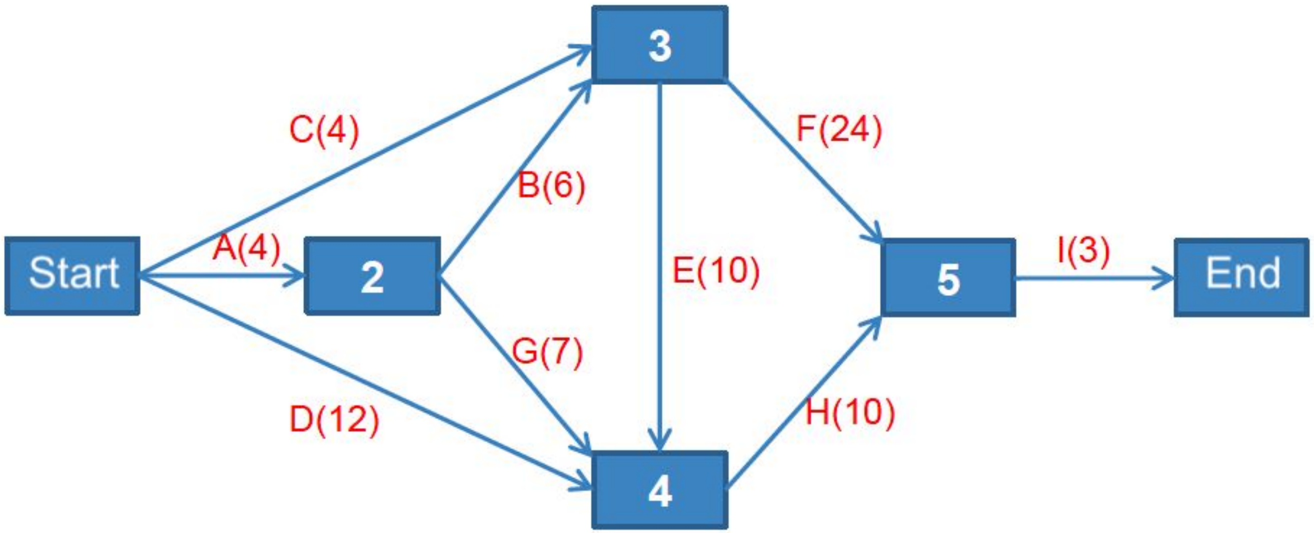
Critical Path Method (CPM)

Example 2: Finding the Critical Path in a Project



Critical Path Method (CPM)

Example 2: Finding the Critical Path in a Project



- CFI: 31
- CEHI: 27
- ABFI: 37
- ABEHI: 33
- AGHI: 24
- DHI: 25

Some Properties of Critical Path

- Critical Path has **zero time flexibility**
- Critical Path is **critical about *"time"***, not about *"difficulty"*, *"quality", etc.*:
  - Some tasks on the "Critical Path" may be easy or trivial
- There could be **several Critical Paths**, if they have the same length
  - **Critical Path can change** if the tasks and their durations change
- Tasks **not** on the Critical Path may have some **slack time**:
  - They can be **delayed *slightly*** without impacting the project deadline
  - But they **cannot delay too much**, otherwise they create the new longest sequence of task, which becomes **the new Critical Path**

Critical Path Analysis

- **Early start (early finish):** the earliest time that the task can be started (or finished), **after predecessor tasks (depended) finished**
- **Late start (late finish):** the latest time that the task can be started (or finished), **before successor tasks (depending) start**
- **Total float:** Amount of time a task **may be delayed from its early start without delaying** the planned **project finish date**
- **Free float:** Amount of time a task **can be delayed without delaying the early start** of any successor tasks

Critical Path Analysis: **Calculation**
→ **Should use AON (PDM) format**

Earliest start: forward

$$t_i = \max_{j \in P(i)} \{t_j + t_{ij}\}$$

Predecessor nodes (before *i*)

Duration of the edge (i, j)

Latest start: backward

$$T_i = \min_{j \in S(i)} \{T_j - t_{ij}\}$$

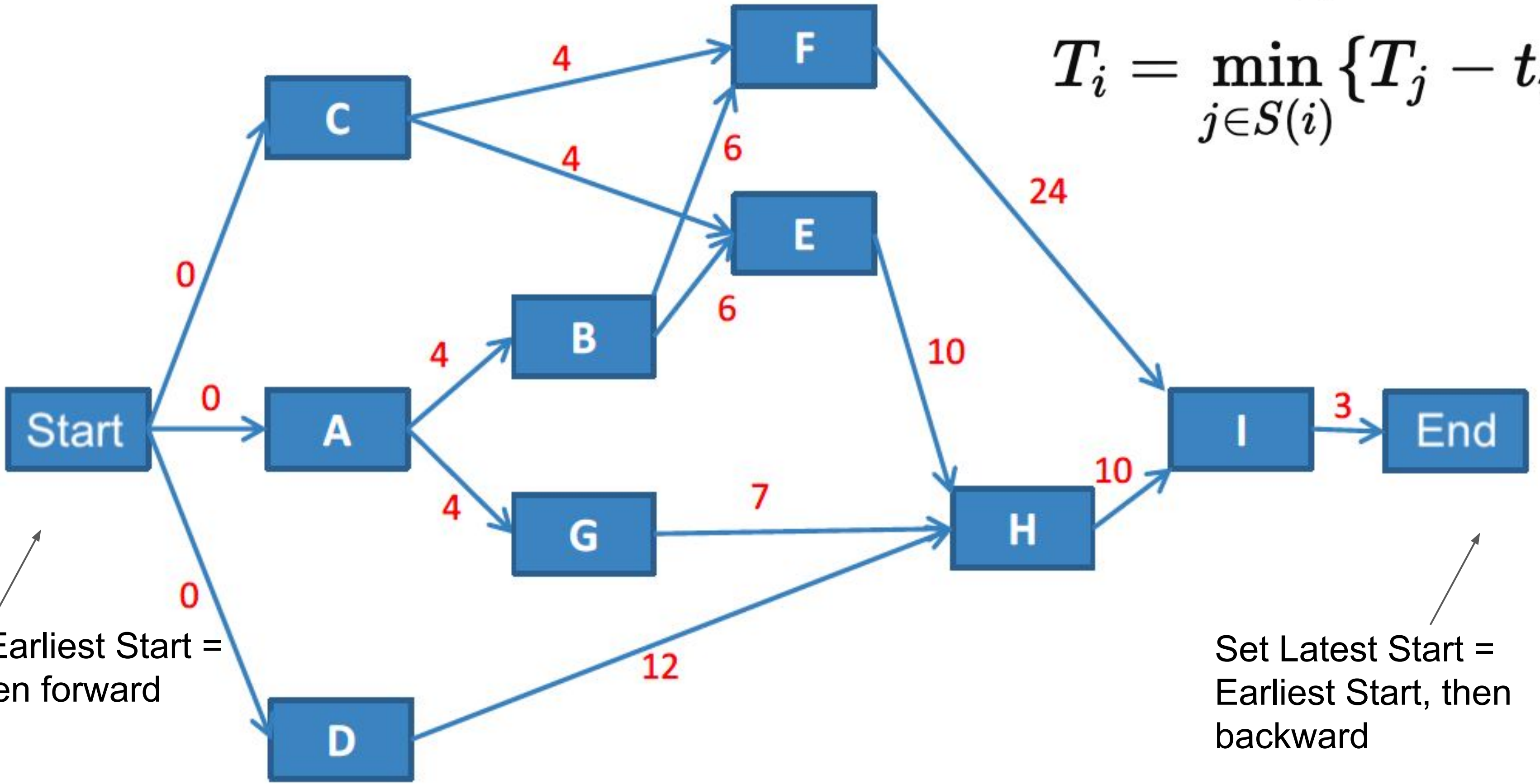
Successor nodes (after *i*)

Duration of the edge (i, j)

Critical Path Analysis: **Example**

$$t_i = \max_{j \in P(i)} \{t_j + t_{ij}\}$$

$$T_i = \min_{j \in S(i)} \{T_j - t_{ij}\}$$



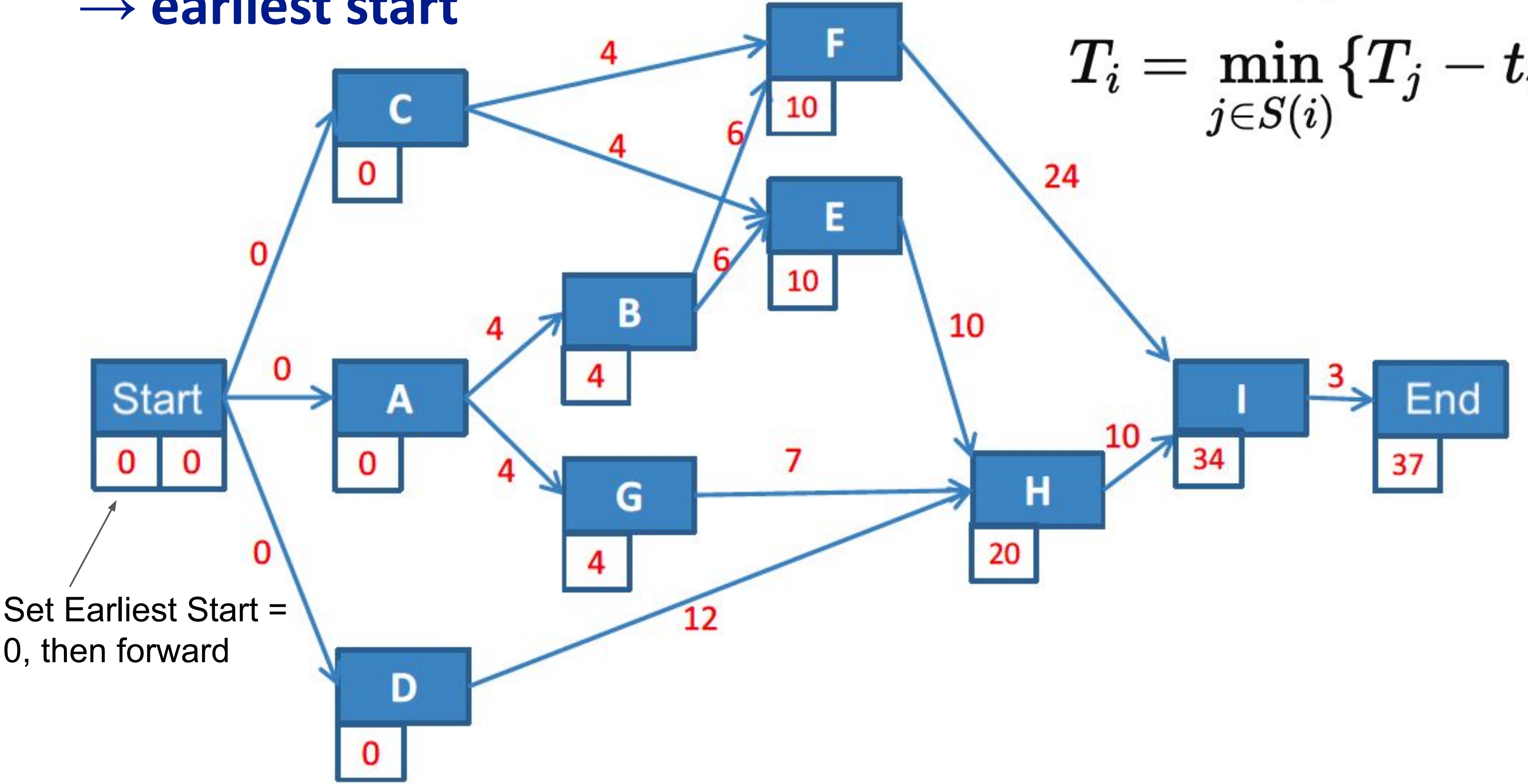
Set Earliest Start = 0, then forward

Set Latest Start = Earliest Start, then backward

Critical Path Analysis: **Example**
→ **earliest start**

$$t_i = \max_{j \in P(i)} \{t_j + t_{ij}\}$$

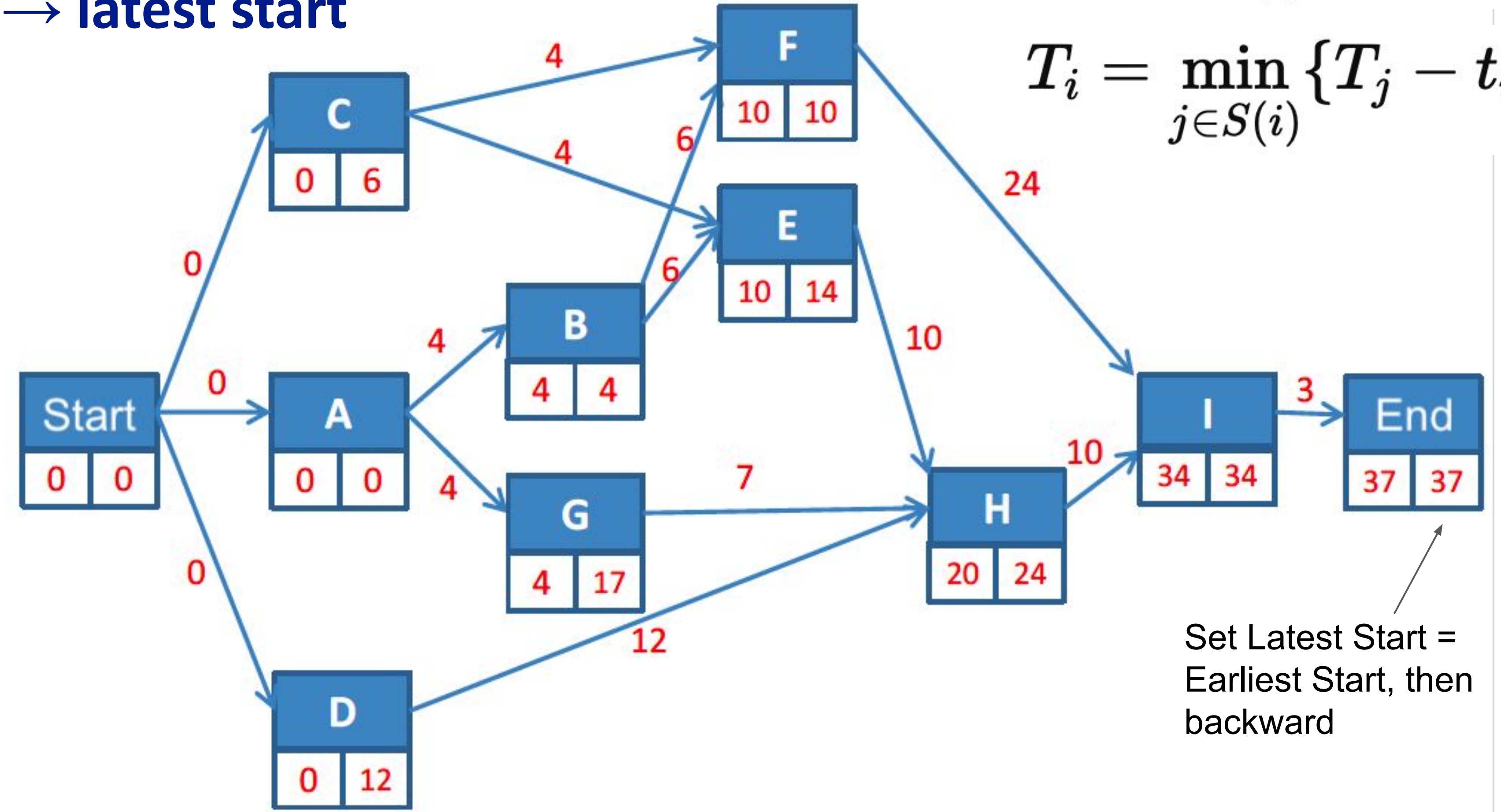
$$T_i = \min_{j \in S(i)} \{T_j - t_{ij}\}$$



Critical Path Analysis: **Example**
$\longrightarrow$ **latest start**

$$t_i = \max_{j \in P(i)} \{t_j + t_{ij}\}$$

$$T_i = \min_{j \in S(i)} \{T_j - t_{ij}\}$$



Set Latest Start = Earliest Start, then backward

Critical Path Analysis: **Calculation**

Total float:

$$M_i = T_i - t_i$$

Free float:

Duration of current node *i* (Length of edge *ij*)

$$m_i = t_j - t_i - t_{ij}$$

Earliest start among the successor nodes (after *i*)

Earliest start of current node *i*

Critical Path Analysis: **Example**
→ **float**

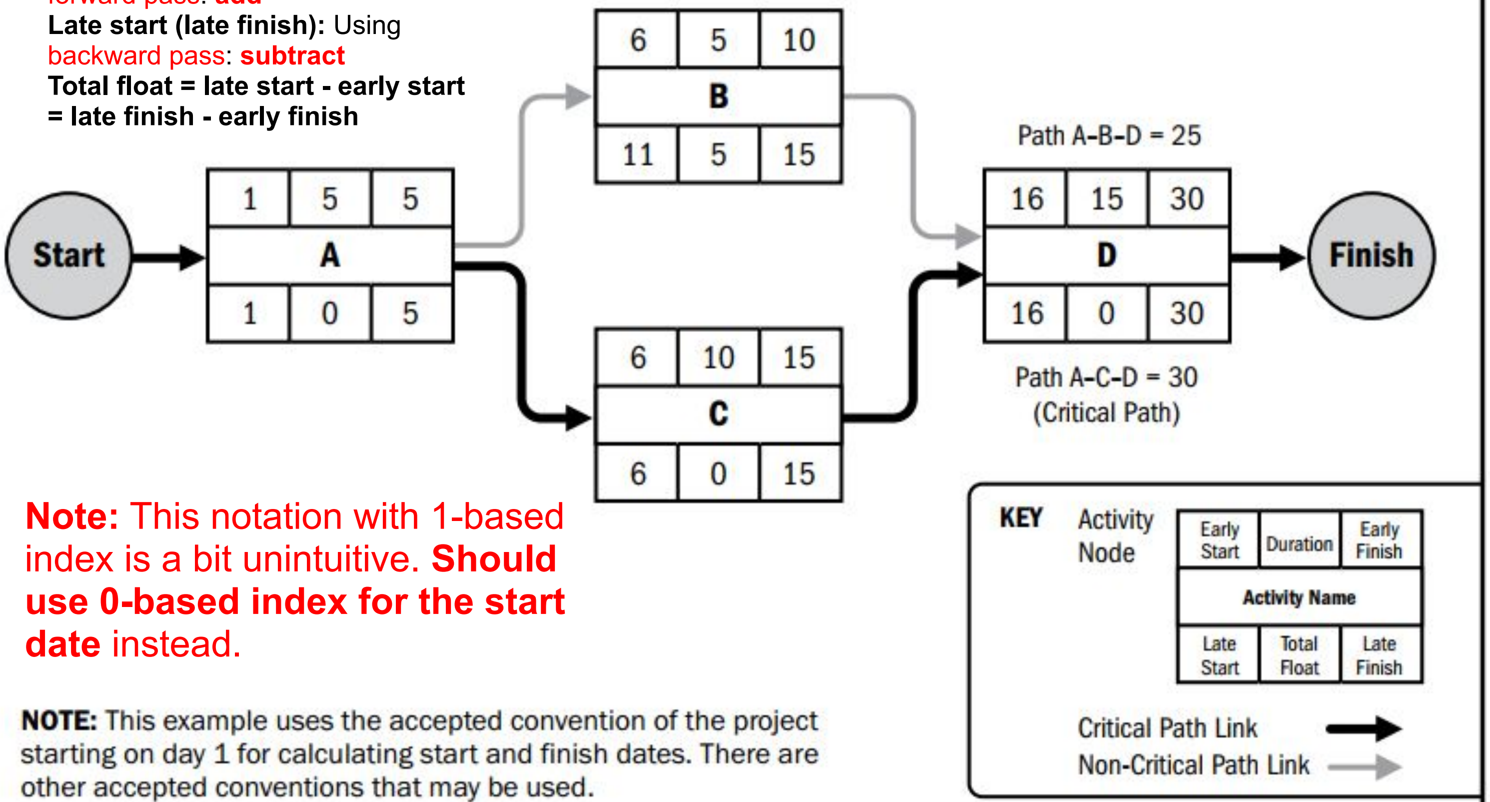
$$M_i = T_i - t_i$$

$$m_i = t_j - t_i - t_{ij}$$

Task	t (Earliest Start Time)	T (Latest Start Time)	Total Float	Free Float
*A	0	0	0	0
*B	4	4	0	0
C	0	6	6	6
D	0	12	12	8
E	10	14	4	0
*F	10	10	0	0
G	4	17	13	9
H	20	24	4	4
*I	34	34	0	0

Critical Path Analysis: **Example (other notations)**

- **Early start (early finish):** Using forward pass: **add**
- **Late start (late finish):** Using backward pass: **subtract**
- **Total float = late start - early start = late finish - early finish**



Note: This notation with 1-based index is a bit unintuitive. **Should use 0-based index for the start date** instead.

NOTE: This example uses the accepted convention of the project starting on day 1 for calculating start and finish dates. There are other accepted conventions that may be used.

Project Scheduling

Modern Techniques

Critical Chain Project Management (CCPM)

What is CCPM?

- **An advanced scheduling technique** that focuses on **resource constraints** instead of just task dependencies
- Developed by **Eliyahu Goldratt** in the 1990s based on the **Theory of Constraints**
- Unlike **CPM**, which assumes unlimited resources, **CCPM assumes limited resources** and ensures realistic scheduling

Critical Chain Project Management (CCPM)

Some concepts:

- **Murphy's Law:** if something can go wrong, it will
- **Parkinson's Law:** work expands to fill the time allowed
- **Student syndrome:** planned procrastination, only begin to work at the last moment before the deadline
- **Limited resources:** resources are limited in practice, tasks using the same resources should not be executed in parallel
- **Human multitasking:** human cannot multitask, they only context-switch between tasks, which is inefficient
- **Buffer:** additional time to complete a task
  - **Project buffer:** additional time added before the project's due date
  - **Feeding buffers:** additional time added before tasks on the critical path

Critical Chain Project Management (CCPM)

Example of **multi-tasking**

that is **bad** (even without accounting for the cost of context-switching)

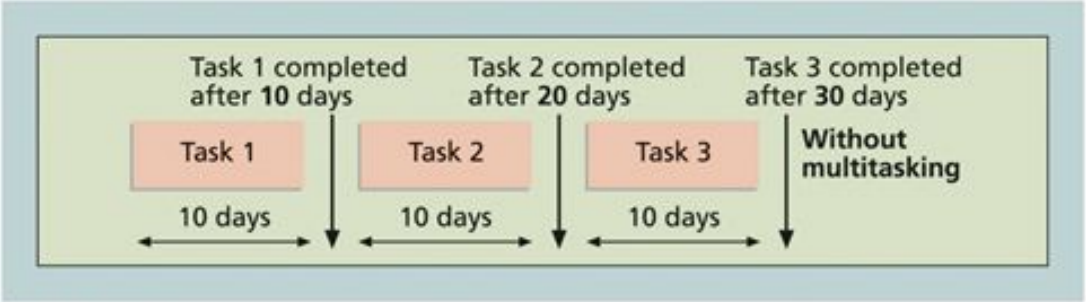


FIGURE 6-10a Three tasks without multitasking

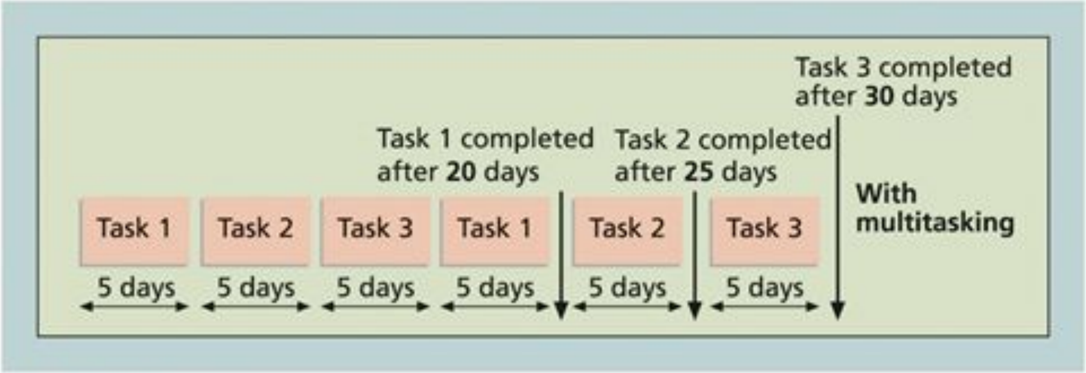


FIGURE 6-10b Three tasks with multitasking

Critical Chain Project Management (CCPM)

Key Ideas of CCPM:

✔ **Buffers:** Extra time is added at the project level **before the deadline**, and **before each resource-sensitive task**

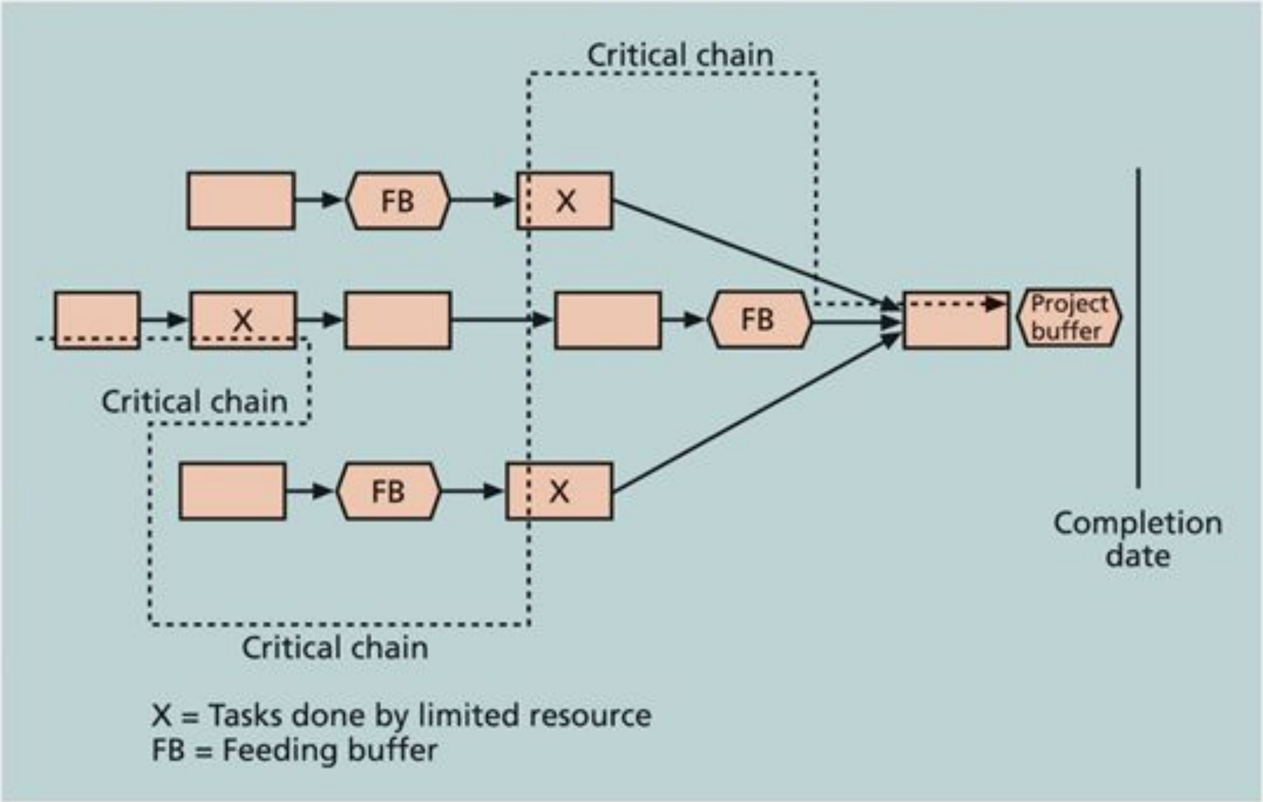
✔ **Resource-based Scheduling:** Tasks are scheduled based on **available resources** and **avoid multitasking**

→ Faster Project Completion: Reduces stress on teams and ensures realistic task durations

→ CCPM resulted in **10% to 50% faster or cheaper** projects than the traditional methods (i.e., CPM, PERT, Gantt, etc.) *

* "Critical Chain Project Management Improves Project Performance". www.pmi.org

Critical Chain Project Management (CCPM)



Source: Eliyahu Goldratt, *Critical Chain*

FIGURE 6-11 Example of critical chain scheduling[8]

Comparison: CPM vs. CCPM

Feature	Critical Path Method	Critical Chain Project Management
Focus	Task dependencies	Resource constraints
Time Buffers	Added to each task	Added at project-level checkpoints
Resource Management	Assumes unlimited resources	Accounts for limited resources
Best For	Creating the initial schedule for a project as a starting point	Optimize the schedule of projects with resource constraints

Example:

- **CPM Approach:** Every individual software module in a project has extra padding (wasted time)
- **CCPM Approach:** Buffers are placed at key checkpoints, improving efficiency

Monte Carlo Simulation for Schedule Risk Analysis

What is Monte Carlo Simulation?

- A **probability-based simulation method** used to predict **best-case, worst-case, and most likely** schedule scenarios
- Helps identify **potential risks and delays** before execution
- Runs **thousands of simulations** to generate a **statistical forecast** of project completion dates
  - How:
    - for each task, assume a distribution, randomly sampling its duration
    - then schedule the project using all tasks' sampled durations

Monte Carlo Simulation for Schedule Risk Analysis

Example: Predicting Website Development Completion Time

- Based on past data, the website project has:
  - Optimistic Time: 8 weeks
  - Most Likely Time: 12 weeks
  - Pessimistic Time: 20 weeks
- Running **10,000 Monte Carlo simulations**, results show:
  - 10% chance of completion in 8 weeks
  - 50% chance of completion in 12 weeks
  - 80% chance of completion in 15 weeks

→ We may have more accurate and detailed estimation

Project Scheduling Optimization and Compression

Measuring Project Performance

Earned Value Management (EVM)

What is EVM?

- A method to **track project cost, schedule, and performance** against the baseline
- Helps **identify deviations early** and take corrective actions
- Combines **scope, time, and cost** to assess project health

→ Details in the next lecture

Techniques to Shorten Project Duration

What is Schedule Compression?

- **Reduce** the **project duration without changing scope**
- Two main techniques:
  - **Fast Tracking** → Try to parallelize tasks
  - **Crashing** → Try to shorten single task

Techniques to Shorten Project Duration

Fast Tracking

- Perform **tasks in parallel** instead of sequentially
- Works **only if activities have discretionary dependencies**
- **Caveat: Fast Tracking** is **low-cost but risky** (rework if errors occur)

Example:

- Normally, **UI design finishes before coding starts**

  with **Fast Tracking**, coding begins **before** the design is fully completed

Techniques to Shorten Project Duration

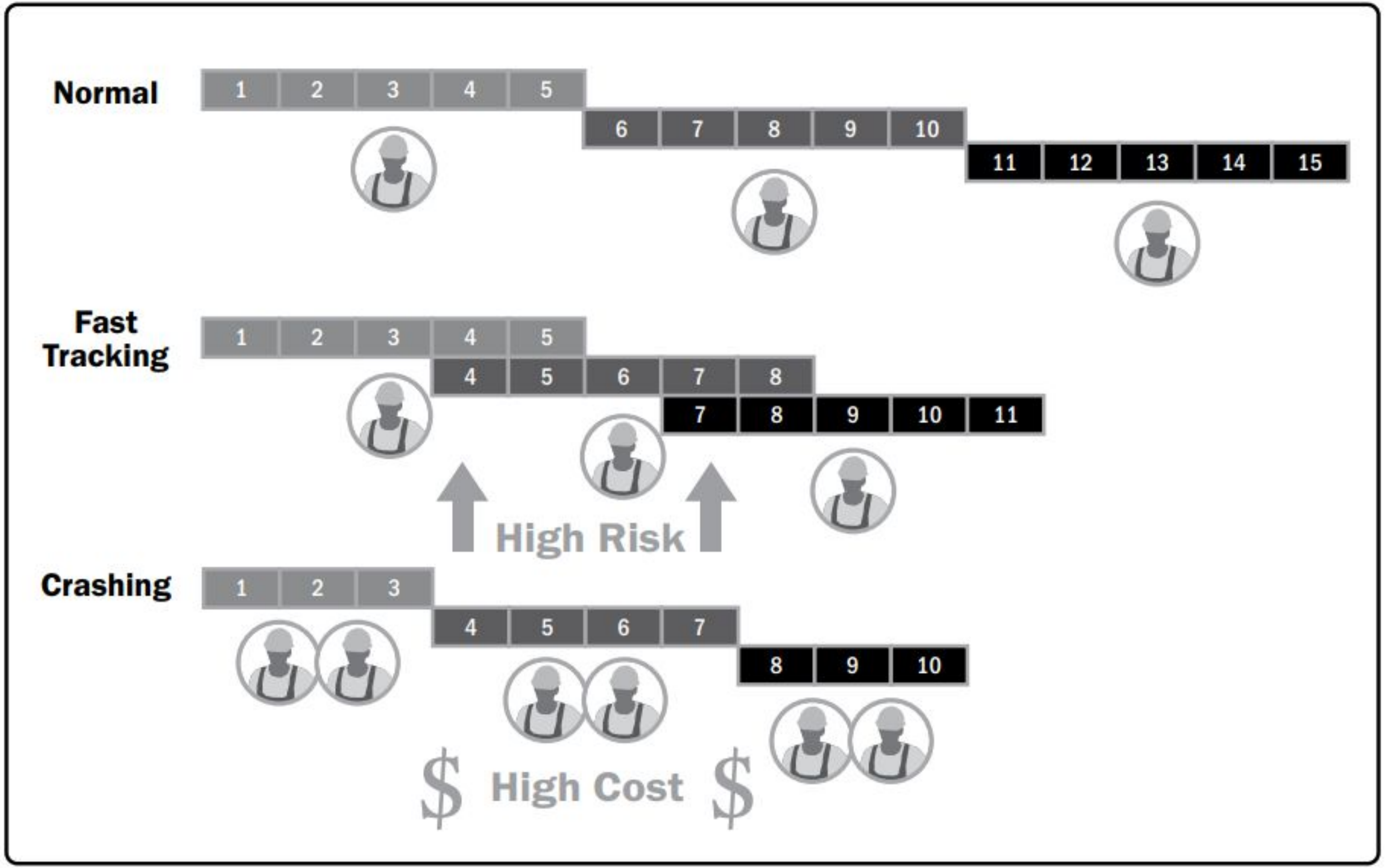
Crashing

- Allocate **extra resources** (overtime, hiring, equipment) to speed up work
- Works **only if adding resources will speed up the process**
- **Caveat: Crashing** is less risky but **costs more money** (extra resources)

Example:

- Hiring **two extra developers** to complete coding faster
- Paying **overtime** for existing developers to work evenings/weekends

Techniques to Shorten Project Duration



Techniques to Shorten Project Duration

Optimizing the Cost when Compressing Schedule

Compressing schedule:

Reduces duration

while **keeping scope**

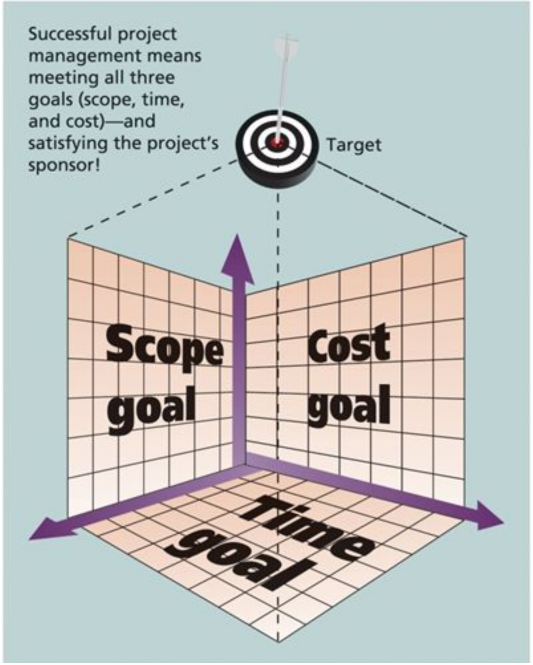
but usually **increases cost**

How to **compress schedule** using **minimal cost**?

→ NP-complete: Exact solutions require exponential time

→ There are heuristics methods: see appendix

→ In practice: use software for approximate solutions



Exercise

Exercise 1

For the given task schedule, do the following:

1. Draw the Network Diagrams in AOA and AON format.
2. Find the Critical Paths (critical tasks and critical path length).
3. Calculate the Earliest Start and Latest Start time of each task.
4. Calculate the Total Float and Free Float of each task.

No.	Task	Predecessor Tasks	Duration
1	A		3
2	B	A	5
3	C	A	3
4	D	B	11
5	E	B	7
6	F	C	4
7	G	E, F	9
8	H	D, G	2

Summary

Summary

📌 You have learned to:

- **Understand project scheduling processes** → Learn how activities are planned, sequenced, and tracked
  - Project scheduling is key to success → Poor planning = missed deadlines
- **Explore key scheduling tools (CPM, CCM, PERT, Gantt, etc.)** → Learn how each method works with real-world examples
  - Use the right tools → Network diagrams, Gantt, CPM, PERT, CCPM, Monte Carlo, etc.
- **Apply time estimation & optimization techniques** → Understand how to estimate task durations accurately and shorten schedules efficiently
  - Plan for flexibility → Measure performance, use buffers, compress schedule

Thank you for listening

Q & A

Appendix

Case study

NASA's Curiosity Rover: Scheduling Challenges

Background:

- **Project:** NASA's *Curiosity Rover* (Mars Science Laboratory)
- **Goal:** Land a **mobile robotic rover on Mars** to explore and analyze the planet's surface
- **Challenge:** Faced **unexpected engineering delays** due to **technical complexities and resource limitations**

Case study

NASA's Curiosity Rover: Scheduling Challenges

Key Scheduling Challenges Faced:

- **Delays in developing the landing system** (new sky-crane technology)
- **Supply chain disruptions** affected hardware availability
- **Testing phases took longer than expected** due to mission-critical risks

Case study

NASA's Curiosity Rover: Scheduling Challenges

Impact:

- Originally planned **for 2009**, the launch was delayed **to 2011**
- NASA needed to **adjust the project schedule** to meet the new launch window (launch window depends on planet orbits → **hard deadlines**)

Case study

How NASA Overcame the Scheduling Challenges

Critical Path Adjustments:

- Identified the **longest sequence of critical tasks** (hardware development, system testing)
- Focused resources on **critical tasks first** to prevent bottlenecks

Case study

How NASA Overcame the Scheduling Challenges

Schedule Buffers:

- **Added time buffers** to critical activities to absorb unexpected delays
- Used a **Monte Carlo Simulation** to predict risks and adjust timelines

Case study

How NASA Overcame the Scheduling Challenges

Fast-Tracking & Parallel Workflows:

- While **hardware components were being manufactured**, NASA started **software testing in parallel**
- Allowed multiple project phases to progress simultaneously

Case study

How NASA Overcame the Scheduling Challenges

Result:

Curiosity Rover successfully launched on November 26, 2011, and landed on Mars on August 6, 2012.

Project Scheduling Optimization and Compression

Techniques to Shorten Project Duration

Optimizing the Cost when Compressing Schedule

A Heuristic Method for Schedule Compression with Minimal Cost:

Step 1: Identify the task on the **critical path** that has the **lowest cost per unit of time reduction**. Reduce its duration as much as possible **until**:

- The task reaches its **minimum feasible duration**, OR
- A **new critical path** appears.

Step 2: Recalculate the earliest start time t_i and latest start time T_i. **Repeat Step 1** if needed.

Step 3 (Multiple Critical Paths):

- If there are **two critical paths**, you **must** reduce time on **both paths** simultaneously.
- Choose **two tasks** (one from each path) that **together have the lowest total cost**.
- Keep adjusting until the total project time is minimized with the **least additional cost**.

Techniques to Shorten Project Duration

Optimizing the Cost when Compressing Schedule: **Example**

How many days can the project be shortened, and what is the minimum cost?

Task	Predecessor Task(s)	Duration (days)	Minimum Duration	Cost per Day Reduced
*A	-	4	2	5
*B	A	6	5	19
C	-	4	2	4
D	-	12	9	10
E	B, C	10	8	5
*F	B, C	24	19	13
G	A	7	6	12
H	D, E, G	10	7	7
*I	F, H	3	2	3

Techniques to Shorten Project Duration

Optimizing the Cost when Compressing Schedule: **Example**

How many days can the project be shortened, and what is the minimum cost?

Crashing method

Reducing critical path activities:

- **Activity I** reduced by **1 day** → cost **3**
- **Activity A** reduced by **2 days** → cost **5 × 2 = 10**
- **Activity F** reduced by **5 days** → cost **13 × 5 = 65**
- **Activity B** reduced by **1 day** → cost **19**

→ **Total cost: 3 + 10 + 65 + 19 = 97**

→ **Current duration: 37 - 1 - 2 - 5 - 1 = 28 days**

Reducing emerging critical path activities:

- **Path ABEHI = 29 days > 28 days**: Need to reduce this path by **1 day**
    - **Reduce Activity E by 1 day** → Cost **5**
- Other paths are already **< 28 days**

Total cost: 97 + 5 = 102

Techniques to Shorten Project Duration

Optimizing the Cost when Compressing Schedule: **Example**

How many days can the project be shortened, and what is the minimum cost?

Crashing Order	ABFI	ABEHI	CFI	CEHI	DHI	AGHI	Crashing Cost
Initial	37	33	31	27	25	24	-
I	36	32	30	26	24	23	3
A	35	31	30	26	24	22	5
A	34	30	30	26	24	21	5
F	33	30	29	26	24	21	13
F	32	30	28	26	24	21	13
F	31	30	27	26	24	21	13
F	30	30	26	26	24	21	13
F + E	29	29	25	25	24	21	5
B	28	28	25	25	24	21	19

Agile and Schedule Management